

BITS Digital CodeForge V1.0

Bug Fix Log – Advanced Grading Console · Darshan Joshi

19 bugs found and fixed. Each fix is marked with a matching `FIX #n` comment in `versions/v1-bugfix.html`, and all fixes are carried into the final `index.html`. Every bug was first reproduced on the original file, then retested on the fixed file using the four Excel test files listed at the end (in the `test-files` folder).

#	Bug / Issue Identified	How You Reproduced It	Root Cause	Fix Implemented	How You Tested the Fix
1	The course dropdown showed the same course many times, and old courses stayed after a new upload	Uploaded Test 1. Each course showed up once per student (108 options). Uploaded Test 2 and the old courses were still there.	The code added one dropdown option for every student row, not for every unique course, and it never cleared the old options.	The dropdown is cleared on every upload, and each course is added only once (sorted).	Test 1 shows only CS F211, MATH F111 and PHY F110. After uploading Test 2, only BIO F110 and ECON F211 are shown.
2	The timer started when the page opened, not when grading started. It also showed nothing for the first second.	Waited on the page without doing anything. The clock was already at 00:33 by the time a course was picked.	The start time was taken when the page loaded, and the clock was only drawn after the first 1-second tick.	The timer starts when a course is selected and draws 00:00 straight away. Switching course starts a new session. The timer pauses on download and continues if ranges are edited again.	The timer stayed at 00:00 until a course was picked, then counted from 00:00.
3	The file picker didn't show .xlsx files	Opened the file picker. Only .xls files could be picked, but the brief asks for .xlsx.	The file input had <code>accept=".xls"</code> .	Changed it to <code>accept=".xlsx,.xls"</code> .	.xlsx files now show up in the picker and load correctly.
4	Min and Max values were swapped	For MATH F111 (marks 0–100), the "Min" box showed 100 and the "Max" box showed 0.	The "Min" label was on the element with id <code>max</code> , and the "Max" label on the one with id <code>min</code> .	Swapped the ids so each label shows its own value.	MATH F111 now shows Min = 0 and Max = 100.
5	A student with a fractional mark disappeared from the results and the CSV	In MATH F111, the student with 79.5 was missing from the grade counts and the downloaded CSV, with no warning. The guidance text also rounded wrongly (80.2 → 81).	79.5 is above A's max (79) and below A's min (80), so it matched no grade and was skipped quietly.	Marks are rounded to the nearest whole number on upload, and the upload message says how many were rounded. Fixed the guidance example (80.4 → 80, 80.5 → 81).	The upload message says "2 fractional mark(s) rounded". 79.5 became 80 and got an A. All 60 MATH F111 students are in the CSV.
6	Cancelling the file picker caused an error	Opened the picker and pressed Cancel. The console showed a "readAsBinaryString... not of type Blob" error.	The code read <code>files[0]</code> without checking that a file was actually picked.	Stops early when no file is picked. Also switched to <code>readAsArrayBuffer</code> (the old method is deprecated) and added error messages for unreadable files.	Cancelled the picker: no console errors.
7	Grade ranges could leave students at the top or bottom without a grade, and download was still allowed	In MATH F111, set A's max to 95. No error appeared, download stayed enabled, and the students with 98 and 100 were missing from the export. Did the same with E's min set to 5.	Validation only checked that ranges joined each other. It never checked that A goes up to 100 and E starts at 0.	Validation now requires A to end at 100 and E to start at 0, and the message names the grade. The summary also shows a red "No grade: n" chip if anyone is left out.	A max 95 shows "A must go up to 100...", download is disabled, and "No grade: 2" appears. E min 5 shows a similar message.

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8	Setting a Min to 0 broke the next grade's Max dropdown	Set A-'s min to 0. B's max dropdown went blank.	The auto-adjust step set B's max to -1, which isn't one of the dropdown options, so the dropdown lost its value.	Auto-adjust stops at 0, and the message explains that there's no room left for lower grades.	A- min 0 leaves B's max at 69 and shows "A- starts at 0, so there is no room left for B..."
9	Stats showed "undefined" and "NaN" when no course was selected	Picked a course, then chose "Select a course to begin" again. Stats showed blank / NaN.	The stats code ran on an empty list and didn't check for zero students.	Going back to the placeholder clears the screen. Stats show "-" when there are no students.	Going back to the placeholder shows "-" in all four boxes, with no errors.
10	The average was a huge wrong number when marks were stored as text in Excel	Picked CS F211 in Test 1 (marks stored as text cells). The average came out as a huge wrong number.	Text marks were joined like strings ("0"+"62"+"71"...) instead of added as numbers.	All marks are converted to numbers on upload.	CS F211 now shows Avg = 69.50, Min 47 and Max 93.
11	The histogram broke with bigger classes and didn't line up with the curve	PHY F110 (40 students, all 85): the bar went off the top of the chart. When all marks were the same, the bell curve didn't draw. The curve also sat slightly to the left of the bars.	Bar height was fixed at 12px per student, so there was no scaling. With identical marks the standard deviation is 0, which caused a divide-by-zero. The curve used 30px per bin while the bars used 32px.	Bars and curve now share one scale that fits the chart. The curve is skipped when every mark is the same. It uses the same spacing as the bars. Each bar shows its count. Bin labels read 0-9, 10-19 ... 90-100. Old animations are cancelled before a new one starts.	PHY F110 shows one bar labelled 40 that fits inside the chart. MATH F111 (60 students) shows the curve sitting on the bars (screenshot checked).
12	The CSV broke on names with commas, had a stray space, and every file had the same name	Instructor "Sharma, Prof. R" split across two columns. The course line read Course, MATH F111 . Every download was called grades.csv .	Values were pasted into the CSV without quoting, and there was a typo in the template.	Values are escaped properly (and can't be read as spreadsheet formulas). Removed the space. The file is named grades_<course>.csv .	The CSV opens with the name in one cell, the course line is correct, and the file is grades_MATH_F111.csv .
13	Reset asked for confirmation twice	Clicked Reset Range and got two confirmation pop-ups in a row.	Two confirm() calls one after the other.	One confirmation only. The Reset button is disabled until a course is selected.	One pop-up appears, and ranges reset to the defaults.
14	The card "lift" and summary "pulse" effects were almost invisible	Changed a range. The card and grade chips barely moved.	The effect class was removed after 30ms, but the CSS transition takes 250ms.	The class is kept for 250ms.	The lift and pulse are now clearly visible when a range changes.
15	The reduced-motion setting still moved the cards	Read the CSS. A second prefers-reduced-motion block added the movement back.	Two conflicting media query blocks.	Removed the second block. The histogram also skips its animation when reduced motion is on.	With reduced motion turned on in the browser, nothing animates.
16	The Excel library version wasn't fixed	The script loaded npm/xlsx with no version, so it always got whatever the latest version was.	No version pin, so a future release could break the app without any code change.	Pinned it to xlsx@0.18.5 .	The file uploads and parses correctly with the pinned version.

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17	You could download with an empty instructor name	Picked a course, cleared the instructor name, then clicked Finalize & Download. The CSV had an empty instructor.	The name was only checked when the course was picked.	The name is checked again at download time, with a message and focus on the field.	An empty name shows "Please enter the instructor name before downloading." and nothing downloads.
18	Wrong ordinal endings in the completion message	Downloaded 21+ times. The message said "21th", "22th" and "23th".	Any number from 4 up just got "th" added.	Proper 1st / 2nd / 3rd / nth logic.	The messages read "21st", "22nd" and "4th" correctly.
19	The file contents were never checked	Uploaded Test 4 (wrong column names): nothing loaded and no message appeared. Uploaded Test 3 (marks of 150, -5 and "abc", blank cells, a repeated BITS ID): everything was accepted.	The code assumed every file was perfect.	The file must have BITS ID, Course and Total Marks (spacing and capitals don't matter). Bad marks, rows with missing values and duplicate IDs are skipped. A message lists which rows were skipped and why.	Test 4 shows "Missing column(s): ...". Test 3 loaded 4 valid students and listed the skipped rows with reasons (rows 6–7 missing values, rows 3–5 bad marks, row 9 duplicate).

Test files used

- **Test_1_Main_Marks.xlsx** – 3 courses, 108 students. Includes fractional marks (79.5, 80.4), edge marks 0 and 100, a course with marks stored as text (CS F211) and a course where 40 students all scored 85 (PHY F110). Covers bugs 1–5, 7–14, 17, 18.
- **Test_2_Second_Upload.xlsx** – different courses, uploaded after Test 1 to check the course list refresh (bug 1) and picker cancel (bug 6).
- **Test_3_Messy_Data.xlsx** – marks of 150, -5 and "abc", missing course, missing marks, a repeated BITS ID (bug 19).
- **Test_4_Wrong_Columns.xlsx** – wrong column names (bug 19).

Each Excel file has a second sheet, "How to test", with step-by-step checks. The console only reads the first sheet.